Nishant Sharma

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Projects

Singularity - AI Deep Research Platform | Mar 2026 – Apr 2026 | singularity.hellonish.dev
FastAPI · ARQ · LangGraph · Qdrant · fastembed/ONNX · Redis · PostgreSQL · Next.js 16 · Docker · AWS · SSE

- Built production multi-agent research pipeline (25,500 LOC) solo in 2 weeks; 6-container Docker stack on AWS t3-small (~$26/month) with: 3 manager agents propose full report trees in parallel, lead synthesizes, every search query targets a planned section.
- Engineered 44-skill auto-registration system (__init_subclass__ metaclass, zero manual wiring) spanning 18 retrieval sources (ArXiv, PubMed, SEC EDGAR, GitHub, ClinicalTrials, YouTube transcripts), 18 analysis skills, 8 output skills; StrengthConfig scales 14 parameters across 3 intensity tiers (21–130+ LLM calls)
- Implemented pure BYOK architecture routing 10 models (xAI Grok, Gemini 2.5 Pro/Flash, DeepSeek R1/V3) through a single injected LLM client; Fernet-encrypted keys at rest, JWT family rotation with reuse detection, 30-second single-use SSE JWT; replaced sentence-transformers with fastembed/ONNX — 1.4GB lighter, no GPU at inference

Wand - AI Career Intelligence Platform | Jan 2026 – Apr 2026 | github
FastAPI · Celery · Redis · WebSockets · Instructor · Pydantic · Gemini · DeepSeek · SQLite · Docker

- Architected 6-step wave-parallelized LLM pipeline (profile extraction, JD parsing, company intel, match scoring, contact strategy, action plan) using asyncio.gather(); ExtractedProfile auto-cached after first run to skip redundant LLM calls; each step streams result via WebSocket in real time as it completes
- Built multi-model weighted scoring engine (Qualification 30%, Technical 25%, Keyword 25%, Formatting 20%) Gemini 2.5 Pro at temperature=0.0 for nuanced qualification reasoning, Flash for pattern-matching all outputs as typed Pydantic models via instructor with zero parsing code across all call sites
- Designed unified LLMClient routing all AI calls across Grok, Gemini, DeepSeek at runtime; cover letter generation across 4 modes (Storyline/Disruptive/Regular/Auto) with JDToneAnalyzer detecting industry and tone signals; multi-source profile unification (PDF + LinkedIn + HTML) with semantic entity alignment and merge-priority rules (Resume > LinkedIn > Portfolio)

Finassistant - Multi-Agent Financial Research Platform | Nov 2025 – Dec 2025 | github
LangGraph · FastAPI · ChromaDB · MongoDB · Gemini 1.5 Flash · yfinance · React 18 · Docker · AWS

- Dual-mode LangGraph system (Chat: 2–4 LLM calls, 2–4s; Think: 10–15 calls, 15–30s) with Planner, Financial Agent, Publisher workflow; 18+ financial tools; SEC filing RAG pipeline (<200ms ChromaDB vector search); MongoDB session persistence; deployed AWS EC2 + S3+CloudFront, 50+ req/sec

Spectral Gradient Equalization - ML Robustness Research | Spring 2026 | NYU Tandon

- Designed novel gradient-level intervention (2D FFT decomposition) to suppress shortcut learning in CNNs without group labels; achieved +3.1pp worst-group accuracy over ERM on Waterbirds benchmark (ResNet-18)

Work Experience

Machine Learning Teaching Assistant | NYU Tandon School of Engineering | NY, USA | Sep 2025 - Present

- Standardized PyTorch training pipelines (data loading, training loops, evaluation) used by 100+ students across course offerings
- Led debugging sessions on CNN optimization, backpropagation, autograd; created video lectures simplifying complex ML calculus concepts for 50+ graduate students

Co-Founder | Macverin Technologies | www.macverin.com | Jul 2022 - Aug 2024

- Architected distributed microservices (FastAPI + Node.js) serving 1K+ DAU for 10+ clients; maintained 99.9% system availability
- Built full-stack products end-to-end: TypeScript/React/Next.js frontends, ML inference backends, AWS deployments, automated CI/CD pipelines

Education

- **New York University - Tandon School of Engineering** | New York, US | Sep 2024 - Expected May 2026
  Master of Science in Computer Engineering · GPA: 3.67
  FOCUS: LANGUAGE MODELS, MULTIMODAL MODELS, TRAINING & FINE-TUNING
- **Dr. A.P.J. Abdul Kalam Technical University** | Delhi, India | Aug 2020 - Jun 2024
  B.Tech Computer Science and Engineering · 1st Division with Distinction

Technical Skills

- **AI/ML:** RAG pipelines, Multi-agent orchestration, LLM fine-tuning (LoRA/QLoRA), RLHF (GRPO/PPO/DPO), Instructor (structured outputs), LangGraph, LangChain, DSPy, Vector stores (Qdrant, Chroma, Pinecone), fastembed/ONNX, TensorRT, MLflow, PyTorch
- **Backend:** FastAPI, PostgreSQL, Redis, ARQ, asyncpg, SQLAlchemy, Celery, WebSockets, SSE, REST, gRPC, Node.js, Django
- **Infra:** Docker, Kubernetes, AWS (EC2, S3, RDS, IAM, CloudFront), Terraform, GitHub Actions, Prometheus, Grafana, Caddy
- **Frontend:** Next.js, React, TypeScript | **Languages:** Python, TypeScript, Java, C++, SQL